

Cauchy's integral formula for derivative
More than 1 singulo



12/27



$$\int_C \frac{f(z)}{(z-z_0)^3} dz = \frac{2\pi i}{(n-1)!} f^{(n-1)}(z_0)$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1$$

Example: $\int_C \frac{\sin^6 z}{(z-\pi/6)^3} dz$ a) $|z|=1$

Step: 1

$$|z|=1$$

$$\Rightarrow |x+iy|=1$$

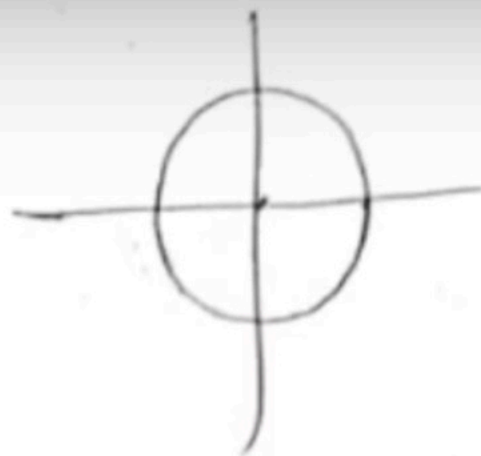
$$\Rightarrow |(x-0) + i(y-0)| = 1$$

center $(0,0)$ radius = 1

Step: 2 Singularities $(z-\pi/6)^3 = 0 \Rightarrow \boxed{z_0 = \pi/6}$

$$\Rightarrow z = \pi/6, \pi/6, \pi/6 \quad \boxed{n=3}$$





$$r_0 = 0.5$$

$$r_0 < 1$$

z_0 is inside the closed curve

$$\int_C \frac{f(z) dz}{(z-z_0)^n} = \int_C \frac{\sin^6 z dz}{(z-r_0)^3}$$

$$f(z) = \sin^6 z = \frac{2\pi i}{(3-1)!} f^{(3-1)}(z_0)$$

$$f'(z) = 6 \sin^5 z \cos z = \frac{2\pi i}{2}$$

$$\begin{aligned} \cos 30 &= \frac{\sqrt{3}}{2} \\ \sin 30 &= \frac{1}{2} \end{aligned}$$

$$f''(z) = 6 \sin^5 z (-\sin z) + \cos z \cdot 30 \cdot \sin^4 z \cdot \cos z$$

$$f''(r_0) = 6 \sin^6(r_0) + 30 \cos(r_0) \cdot \sin^4(r_0)$$

$$= 6 \cdot 0.09 + 30 \cdot 0.999$$

$$= 15.82$$

$$f''(z) \Rightarrow 6 \sin^4\left(\frac{\sqrt{3}}{2}\right) + 30 \cos^2\left(\frac{1}{2}\right) \cdot \sin^4\left(\frac{\sqrt{3}}{2}\right)$$

$$= \underline{\underline{-6 \times 27}} \quad \left(\frac{21}{16}\right) = f''(\pi/6)$$

$$\left(\frac{21\pi i}{16}\right)$$



Different singularities

$$\int_C \frac{f(z) dz}{(z-z_1)(z-z_2)}$$

Both are inside

step 1 $\int(\quad) + \int(\quad)$

2 $2\pi i f(z_0)$